



Future trends

Emerging mobile security paradigms and tech

The future of mobile security will be defined by AI-driven threat detection, zero-trust architectures, and post-quantum cryptography, all operating within increasingly complex 5G and IoT ecosystems. Behavioural and continuous authentication methods will reduce friction while improving risk assessment, and decentralized identity systems – leveraging passkeys and blockchain – will supplant legacy passwords.

Hardware advances such as secure enclaves, Ultra-Wideband (UWB) proximity authentication, and edge-based security services will further harden devices against evolving attacks. Throughout this chapter, we outline key trends, offer step-by-step implementation guidance, and present real-world examples so you can start preparing your mobile defenses today.

AI/ML-driven threat detection and response

Artificial intelligence and machine learning will power next-

generation on-device security agents, identifying anomalous behavior in real time without constant cloud dependence. These models will detect zero-click malware – malicious code that executes without user interaction – by analyzing memory patterns and unexpected API calls.

1. **Deploy On-Device ML Agents:** Integrate lightweight ML frameworks (e.g., TensorFlow Lite) into critical apps to monitor process behavior locally.
2. **Train Adaptive Models:** Regularly update models via federated learning, aggregating threat data from devices without exposing raw logs.
3. **Automate Response:** Configure the agent to quarantine or sandbox suspicious apps upon detection, triggering alerts to the user or enterprise SIEM.

Example: Samsung's One UI 8 beta includes "Advanced Protection" features powered by on-device ML, blocking novel malware strains before signature updates arrive.